

IP Security:

It is a framework for protecting communication over IP.

Applⁿ of IpSec:

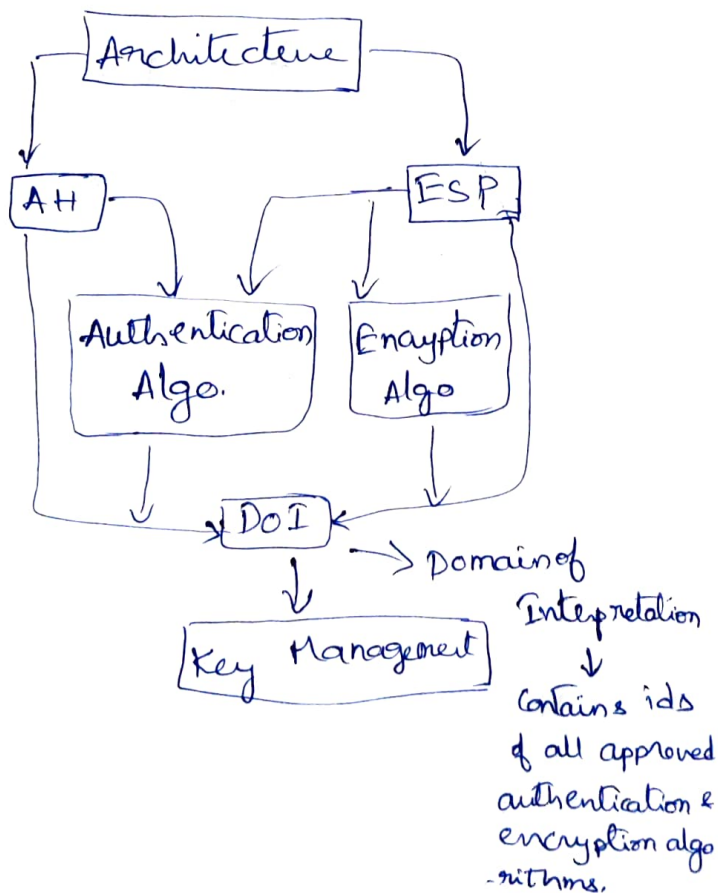
- Secure branch office connectivity over internet.
- Secure remote access over internet.
- Enhancing electronic Commerce security

IP Security Architecture:

- combination of (2) protocols

Authentication
header
(AH)

Encapsulating
security payload,
(ESP)



Authentication Header: (AH)

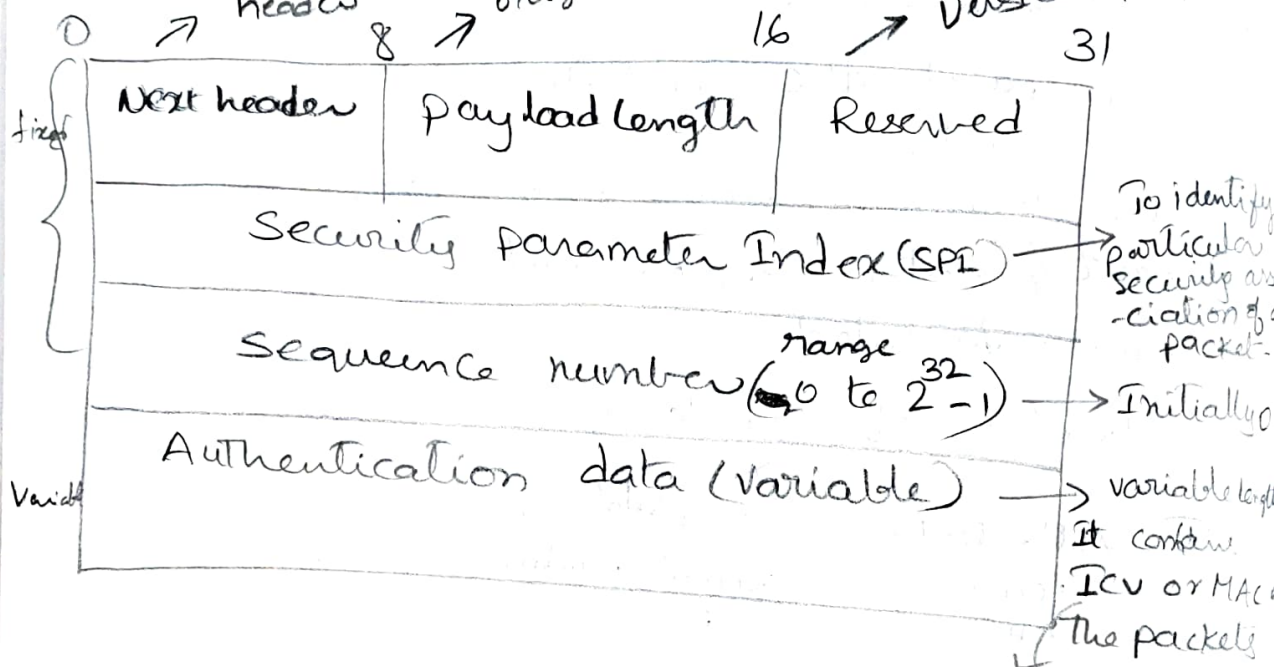
for integrity & authentication

(0-31 bits)

Tells type of next header

length of original data

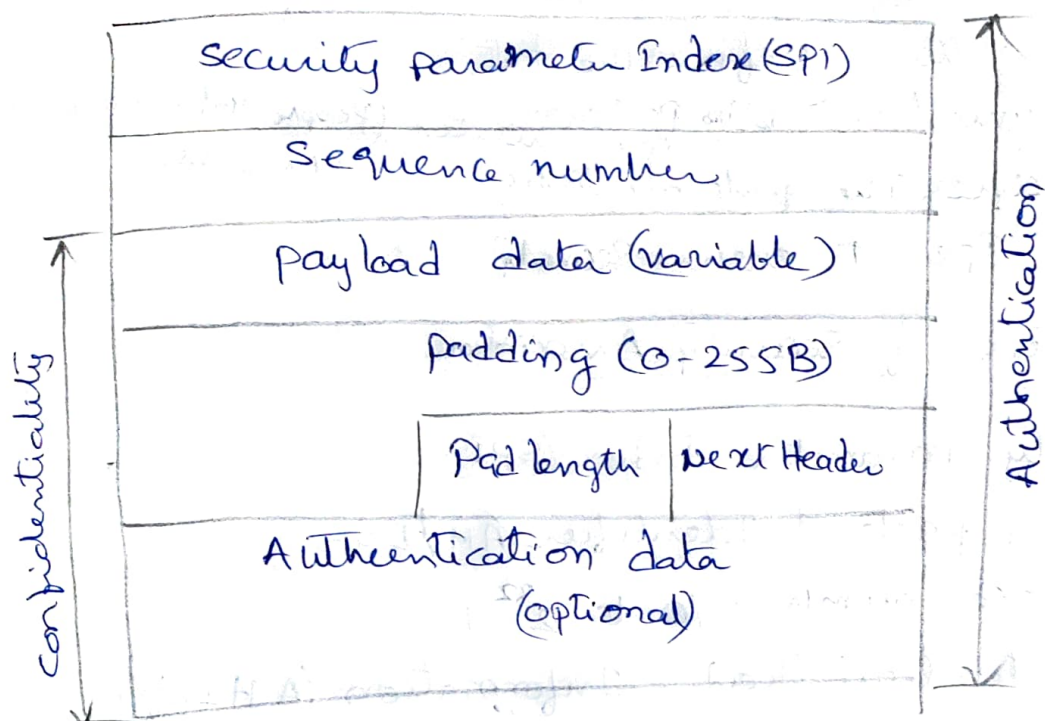
future versions purpose



Integrity check value

↓
If any unwanted changes ^{done} occur in the transmission, that will be determined by this value.

Encapsulating Security Payload: (Esp)



Security Associations: (SA)

- It is a contract shared b/w all the entities before the start of communication. $\rightarrow$ ~~H~~ participating in communication (people, gateways, Routers, LANS etc...)
- SA specifies the protocols to be used in IPsec to ensure security

Parameters of Security Associations:

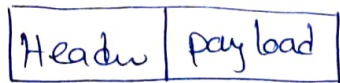
1. Security Parameter Index (SPI)
2. Security protocol Identifier (AH/ESP)
3. Sequence number (0 to $2^{32}-1$)
4. Authentication Header Information (AH Info)
5. ESP Info. (what are the protocols, algos, keys being used)
6. Life time of a Security Association (valid upto)
7. IPsec protocol mode $\rightarrow$ 2 modes
 / \
Transport mode Tunnel mode.

Transport mode: In this mode

payload $\rightarrow$ encrypted

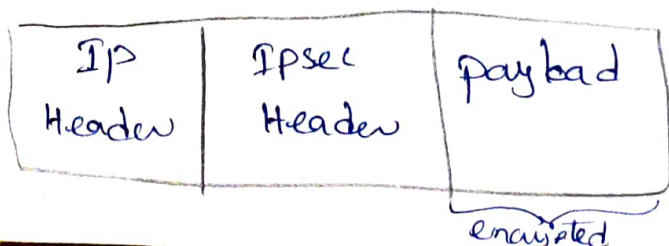
Header $\rightarrow$ not encrypted

Initially,



$\rightarrow$ packet

later, in transport mode, we inter insert IPsec header in b/w



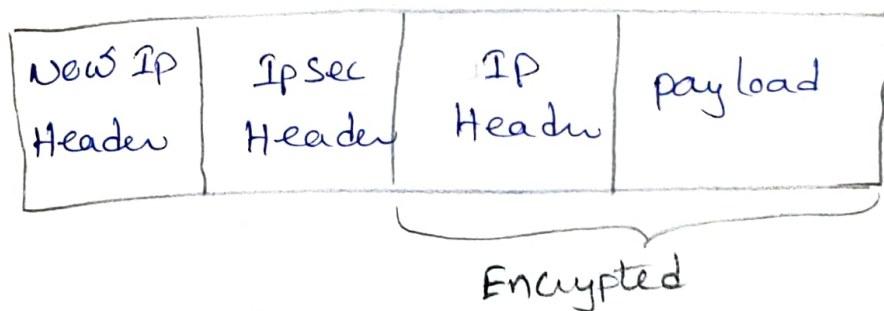
what ever data present after Ipsec header will be encrypted, before it will not encrypted.

In this direct host to host communication happens.

Tunnel mode:

Both payload } Both are
Header } encrypted.

ie. entire pkt is encrypted as result, new IP header is generated).

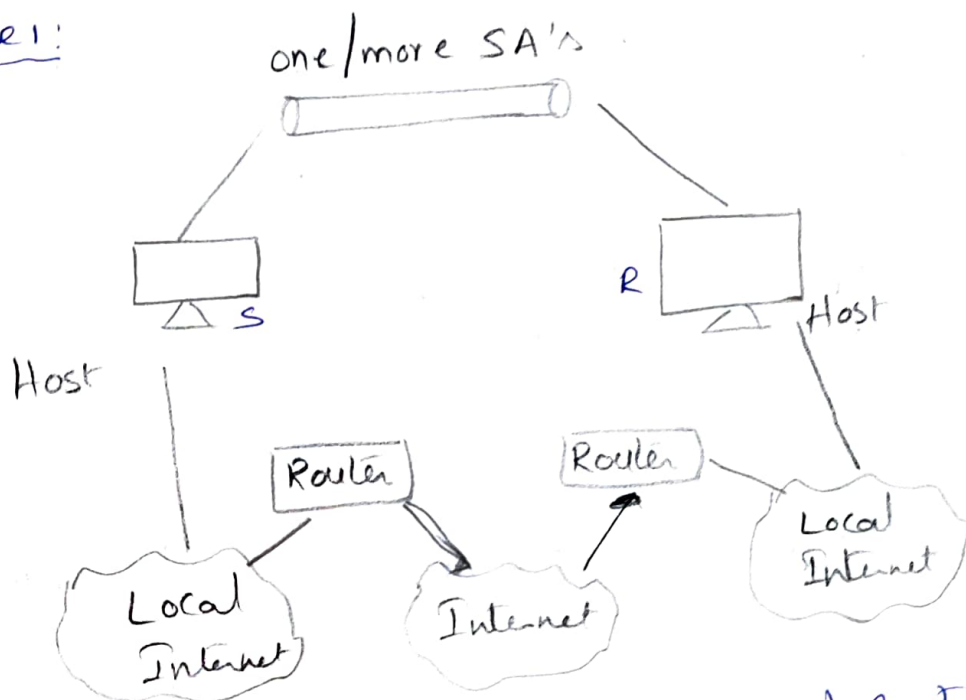


gateway to gateway communication is done.

Combining Security Associations:

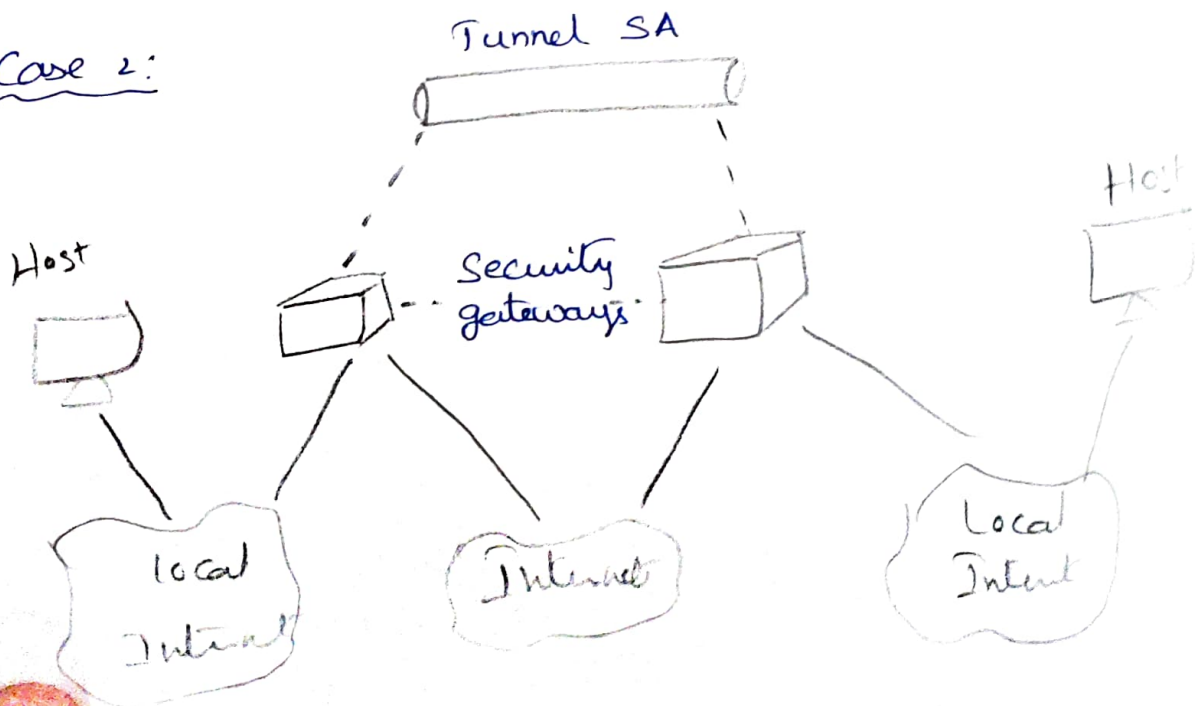
- with individual SA's we can implement either AH/ESP but not both.
- so when both are required, we need to combine multiple SA's.
- There are 4 combinations.

Case 1:



- providing security to the end systems bcz the host is directly connected to SA's.

Case 2:



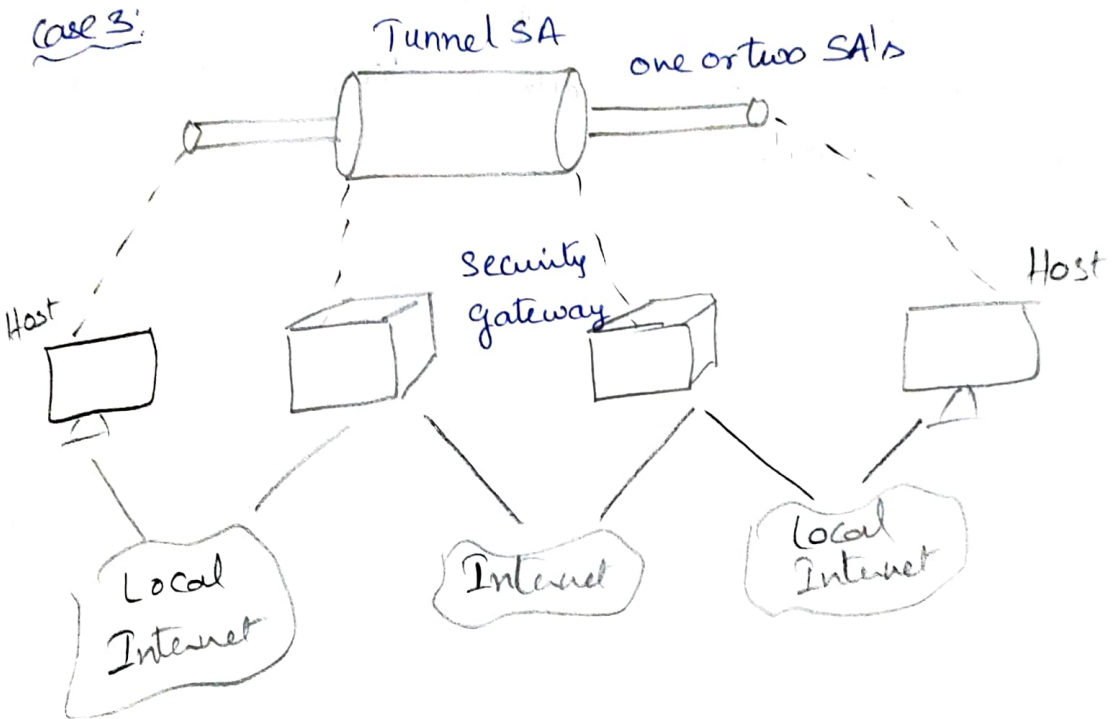
we use Single Tunnel SA.

But, here SA is connected to gateways

mostly used in VPN.

connected to host
(via local Internet)

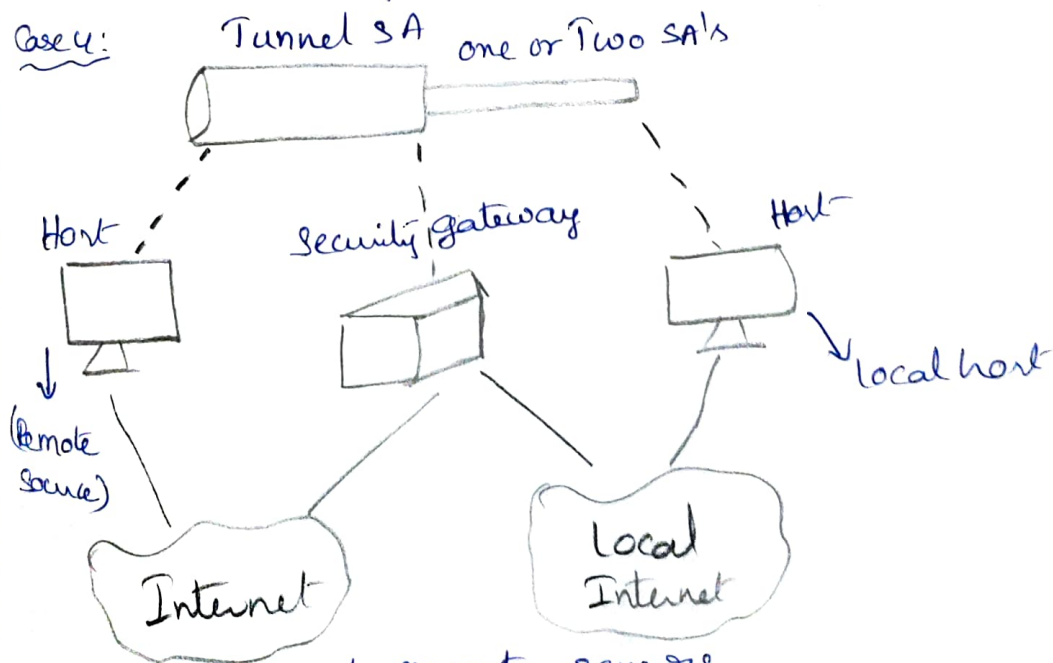
Case 3:



Combination of Case 1 and Case 2.

Tunnel Security for gateways + SA to end to end system also.

Case 4:



Used in case of remote sensors.

Blw remote host & gateway → Tunnel mode

Blw " & local host → one or two SA's.